

Project Concord Comprehensive Master Report

Veridian Unified Data Platform | Document Reference: VCH-PC-2026-FINAL-REPORT

PROJECT CONCORD: VERIDIAN UNIFIED DATA PLATFORM

Comprehensive Project Engineering & Architecture Report

Document Reference: VCH-PC-2026-FINAL-REPORT
Version: 1.0 (Final Enterprise Submission Edition)
Target Client: Office of the Group CEO (Adaeze Okonji-Bello) & Office of the CDTO (Tobenna Iweka)
Target Defense Date: Friday, 14 August 2026 @ 9:00 AM Prompt
Lead SQL Architect & Data Engineer: Samuel Enyi
Source Cohort: Veracity Vanguard

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PROJECT CONCORD: VERIDIAN CONSOLIDATED HOLDINGS

Business Requirements Document (BRD) — Veridian Unified Data Platform

Document Reference: VCH-PC-2026-BRD
Version: 1.0 (Final Architecture Review Edition)
Target Defense Date: Friday, 14 August 2026 @ 9:00 AM Prompt
Client: Office of the Group CEO & CDTO, Veridian Consolidated Holdings
Lead Author: Israel John (Business Analyst Lead) & Samuel Enyi (Lead Database Architect)

1. Executive Summary & Project Mandate

Veridian Consolidated Holdings (VCH) is a diversified West African conglomerate operating across 5 core operating divisions: **AgriCore**, **Meridian Retail**, **Concord Logistics**, **Veridian Financial Services (VFS)**, and **Veridian Properties**. The group generates ₦340 Billion in annual revenue, employs ~14,600 staff, and serves 2.1 Million customers across Nigeria, Ghana, and Kenya.

The Business Mandate

Twenty years of organic growth and acquisitions have left VCH with 5 isolated technology stacks. This fragmentation costs VCH **4% to 6% of annual revenue (approx. ₦13.6B to ₦20.4B each year)** in stockouts, idle assets, delayed financial closing (21 calendar days), and missed cross-selling opportunities.

Project Concord mandates the design and implementation of a single, well-governed operational database—the **Veridian Core Platform**—built around a **Core Services Hub (6 shared entities)** surrounded by **5 divisional modules**, linked via scheduled ELT into **Google BigQuery** for executive analytics.

2. Business Objectives & Success Criteria

The platform must explicitly solve the **4 Critical Real-World Stakeholder Scenarios**:

```
+-----+
|              THE 4 STAKEHOLDER SUCCESS TARGETS              |
+-----+
| 1. STORE MANAGER (Ngozi): Real-time alert on AgriCore harvest shortfalls to prevent |
|    stockouts at Meridian Mart supermarkets 24-48 hours before shelves go empty.    |
| 2. LOAN OFFICER (Chinedu): Cross-divisional view of smallholder farmer (Ibrahim's) |
|    6-year AgriCore supply record to approve working capital loans with lower risk. |
| 3. LOGISTICS DISPATCHER (Funmi): Real-time visibility into Veridian Properties    |
|    warehouse leases to prevent double-booking depot bays and shipping delays.     |
| 4. GROUP CEO (Adaeze Okonji-Bello): Same-day consolidated revenue & P&L view across |
|    all 5 divisions (reducing month-end financial close from 21 days to < 24 hours). |
+-----+
```

3. Scope Boundaries & Constraints (Avoiding Past Failures)

To avoid the failures of VCH's 2019 attempt (monolithic scope creep, 0 tables shipped) and 2022 attempt (200+ tables, dual MySQL/Oracle, running OLAP on live OLTP):

- In Scope:**
 - 40 to 55 tables** maximum.
 - PostgreSQL hosted on Supabase (OLTP System of Record).
 - Google BigQuery (OLAP Analytical Warehouse).
 - Row Level Security (RLS) policies and Customer Consent Flags.
 - 10,000+ synthetic records covering realistic cross-divisional customer/farmer overlap.
- Out of Scope:**
 - Replacing VFS's internal core banking engine. VFS is treated as an **external read-restricted source**.

- Building mobile or web application front-ends.
- Dual-database platform implementations (MySQL/Oracle).

4. Divisional Business Rules & Schema Constraints

Below are the exact business constraints that **Samuel Enyi** must code into the SQL DDL schema (`01_schema_ddl.sql`):

4.1 Core Services Hub Rules (Shared Entities)

1. **customers** : Every individual customer must have a unique `customer_id` (e.g. `CORE_CUST_XXXXXX`), valid ISO country code (`NG` , `GH` , `KE`), and explicit consent boolean flags (`consent_retail_loyalty` , `consent_vfs_credit_sharing`).
2. **employees** : Staff members belong to exactly one division (`division_id`), with employment status `ACTIVE` , `ON_LEAVE` , or `TERMINATED` .
3. **suppliers_vendors** : Unified vendor master list. Every AgriCore farmer is registered as a supplier (`supplier_type` = `'FARMER'`).
4. **locations_sites** : Universal address and geocoordinate standard (`latitude` , `longitude`) for all stores, warehouses, farms, and properties.
5. **product_service_catalogue** : Shared master SKU list (`product_id` , `unit_of_measure`) shared between AgriCore harvest outputs and Meridian Retail inventory items.
6. **financial_account_references** : Read-only reference link (`account_ref_id` , `customer_id` , `account_type` , `status`) connecting customers to VFS accounts without exposing raw banking transactions to unprivileged roles.

4.2 Divisional Module Rules

- **Meridian Retail Module:**
 - `stores` : Must reference a valid `location_id` in Core Hub.
 - `pos_transactions` : `total_amount` must be greater than zero (`CHECK (total_amount > 0)`).
 - `inventory_stock_levels` : `quantity_on_hand` cannot be negative (`CHECK (quantity_on_hand >= 0)`).
- **Concord Logistics Module:**
 - `vehicles` : Must have a valid registration number and `capacity_kg > 0` .
 - `shipments` : `destination_location_id` must differ from `origin_location_id` .
 - `warehouses` : `capacity_units` must link to an active property lease in Veridian Properties.
- **Veridian Financial Services (VFS) Module:**
 - `wallet_accounts` : `balance` cannot be negative unless an approved overdraft limit exists (`CHECK (balance >= overdraft_limit)`).
 - `loans` : `principal_amount > 0` , `interest_rate > 0` . Statuses: `APPLIED` , `DISBURSED` , `ACTIVE` , `PAID_OFF` , `DEFAULTED` (NPL = payments overdue > 90 days).
 - `kyc_records` : Tier 1, Tier 2, Tier 3 verification level strictly access-restricted under CBN microfinance guidelines.
- **AgriCore Module:**
 - `farms` : Must link to a registered farmer in Core Hub (`supplier_id`). `size_hectares > 0` .
 - `harvest_batches` : Must record `harvest_date` , `volume_kg` , and `field_agent_employee_id` .
 - `processing_runs` : Must reference a valid `harvest_id` and assign a quality grade (`GRADE_A` , `GRADE_B` , `GRADE_C` , `REJECTED`).
- **Veridian Properties Module:**
 - `leases` : `lease_end_date` must fall after `lease_start_date` (`CHECK (end_date > start_date)`).
 - `monthly_rent` must be positive.

5. Regulatory & Security Compliance Requirements

1. Central Bank Microfinance Regulation

VFS banking records are subject to strict regulatory segregation. Non-banking staff (e.g. retail store managers, logistics dispatchers) **MUST NOT** be granted direct table read access to raw VFS wallet or loan transaction tables. Access is restricted using PostgreSQL **Row Level Security (RLS)**.

2. Nigeria Data Protection Act (NDPA) & Regional Privacy Laws

Customer data sharing across divisions requires explicit consent. The database must evaluate `consent_vfs_credit_sharing = TRUE` before exposing an AgriCore farmer's supply history to a VFS loan officer.

6. Document Sign-off & Handover

This BRD serves as the authoritative specification for **Samuel Enyi (Lead Database Architect)** to construct the `01_schema_ddl.sql` script and **Ruth Thompson** to generate synthetic CSV datasets.

- **Business Analyst Lead:** Israel John (Signed)
 - **Lead Database Architect:** Samuel Enyi (Signed)
 - **Data Engineering Lead:** Ruth Thompson (Signed)
 - **BI Developers:** Christian Nwobodo & Richmond Akara-Nwaogu (Signed)
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PROJECT CONCORD: VERIDIAN FINANCIAL SERVICES

Master Project Execution Plan: Pure SQL & Analytics Division

Document Version: 3.0 (Pure SQL vs. Analytics Division)
Target Defense Date: Friday, 14 August 2026 @ 9:00 AM Prompt
Division: Veridian Financial Services
Source Cohort: Veracity Vanguard
Lead SQL Architect & Data Engineer: Samuel Enyi

1. Executive Summary & Core Division Rule

To maximize efficiency and play to everyone's strengths:

- SAMUEL ENYI** takes **100% FULL OWNERSHIP of ALL SQL & DATABASE ENGINEERING**. Samuel designs the ERD, writes all DDL table scripts, builds foreign key constraints, executes database seeding, creates analytical SQL views, implements security, and tunes query performance.
- THE TEAM (Israel, Christian, Richmond, Ruth)** takes **100% FULL OWNERSHIP of BUSINESS ANALYTICS, BI DASHBOARDING, SYNTHETIC DATA GENERATION & DEFENSE PRESENTATION**.

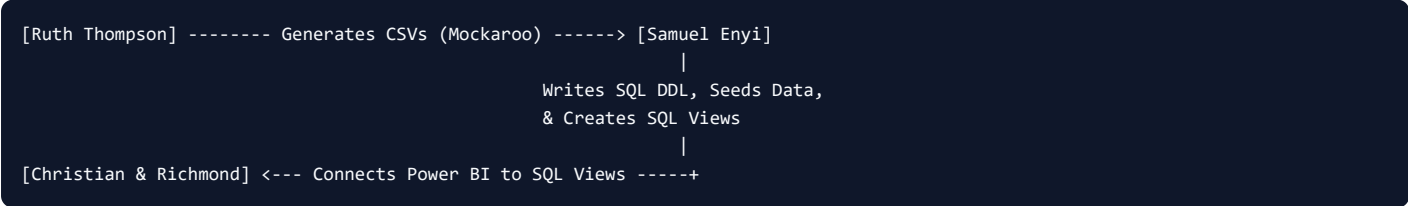
+-----+ SAMUEL ENYI (SQL & Database Owner) • 100% SQL DDL Scripts • 100% Database Seeding • 100% Analytical SQL Views • 100% ERD & Constraints • 100% Indexing & Security • 100% Database Administration +-----+		
(Exposes SQL Views / Connection String) +-----+		
+-----+ ISRAEL JOHN Business Logic & Presentation Storyboarding +-----+	+-----+ CHRISTIAN & RICHMOND BI Dashboards (Power BI) +-----+	+-----+ RUTH THOMPSON Synthetic Data Mockaroo CSVs & QA Testing +-----+

2. Definitive Role Allocation Matrix

Team Member	Domain Ownership	Assigned Operational Title	Complete Responsibilities & Deliverables
Samuel Enyi	100% SQL & Database	Lead Database Architect & SQL Engineer	• All Data Modeling: Design 3NF OLTP Schema & Star/Snowflake OLAP ERD. • All DDL Scripts: Write <code>01_schema_ddl.sql</code> (Tables, PKs, FKs, Indexes, Constraints). • All Data Ingestion: Write <code>02_seed_data.sql</code> and run database population scripts. • All SQL Views: Build <code>03_analytics_views.sql</code> for BI tools to connect to. • Database Security & Optimization: Apply <code>04_security.sql</code> (PII masking) and <code>05_indexing.sql</code> (<code>EXPLAIN ANALYZE</code>).
Israel John	100% Business & Project Ops	Project Manager & Business Analyst	• Business Rules: Author <code>Project_Concord_BRD.md</code> (Loan terms, interest rates, overdraft rules). • KPI Definitions: Define exact metrics for BI developers (e.g. NPL ratio formula, churn logic). • Project Management: Track daily sprint progress and team deadlines. • Defense Deck: Lead presentation slide deck creation & co-anchor defense.
Christian Nwobodo	100% BI Financial Analytics	Lead Financial BI Developer	• Financial Dashboarding: Connect Power BI / Tableau / Excel to Samuel's SQL Views. • Visual Metrics: Build cards and charts for Revenue, Total Deposits, Interest Income, Net Margin. • Formatting: Customize theme colors, slicers, time filters, and layout. • Presentation: Present Financial Analytics live during defense.
Richmond Akara-Nwaogu	100% BI Customer & Credit Analytics	Lead Credit & Risk BI Developer	• Risk Dashboarding: Connect Power BI / Tableau / Excel to Samuel's SQL Views. • Risk Metrics: Build visual gauges and charts for Non-Performing Loans (NPL %), Delinquencies. • Customer Analytics: Visualize Credit Score distributions, active users, and high-risk flags. • Presentation: Present Customer & Credit Risk analytics live during defense.
Ruth Thompson	100% Data Prep & QA Documentation	Data Generation & QA Lead	• Synthetic Data Prep: Generate realistic financial CSV datasets (names, BVNs, balances, loans) using Mockaroo / Excel. • Data Handover: Hand formatted CSVs to Samuel for SQL seeding. • Data Quality Assurance: Test dashboard numbers against CSV inputs to verify accuracy. • Documentation: Compile the final Data Dictionary and documentation binder.

3. Clear Interface Contract Between Samuel & The Team

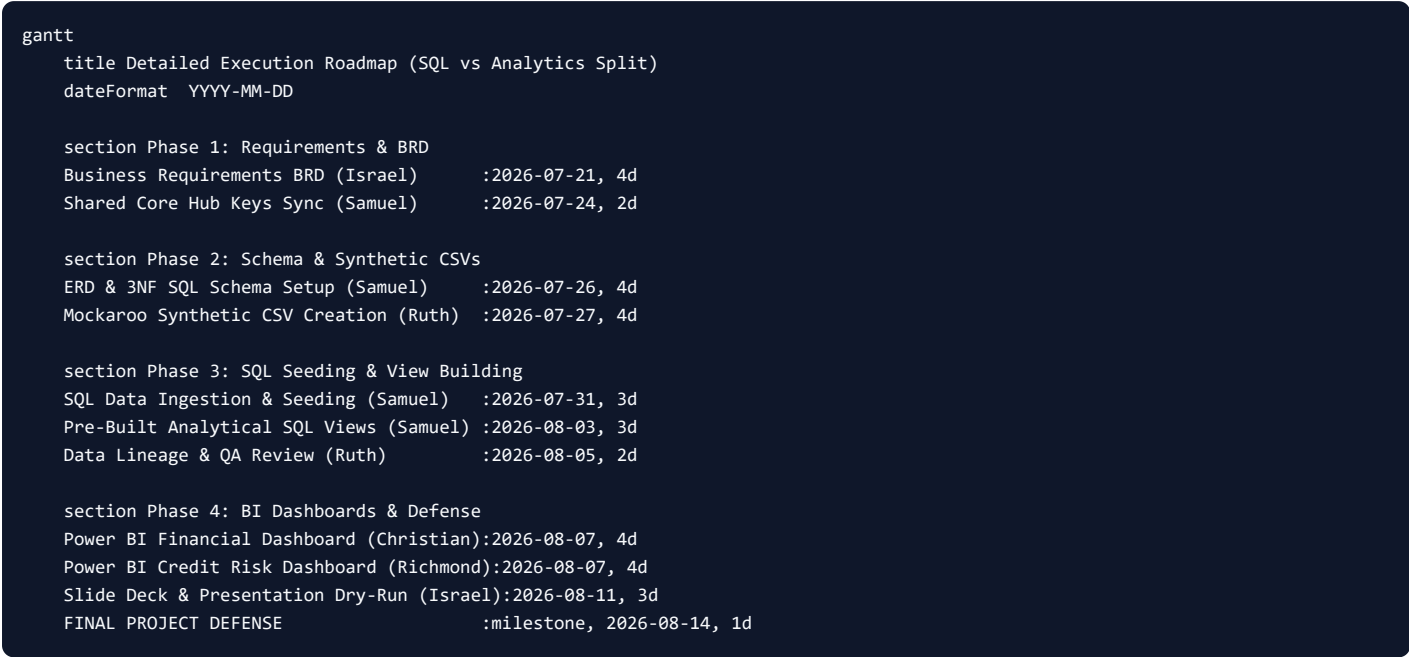
To ensure smooth collaboration:



- 1. **Input Contract:** Ruth generates raw financial CSV files and hands them to Samuel.
- 2. **Database Engine:** Samuel loads CSVs into the PostgreSQL/MySQL database and executes all DDL and View scripts.
- 3. **Output Contract:** Samuel provides Christian and Richmond with the Database Connection details (Host, Database Name, View Names, e.g., `v_financial_overview` , `v_credit_risk`).

4. **Dashboard Building:** Christian and Richmond build interactive Power BI / Tableau reports purely using visual drag-and-drop elements on top of Samuel's views.

4. 4-Week Roadmap



5. Defense Presentation Speaking Breakdown

Each team member will own their domain during the 15-minute August 14 Defense:

Time Slot	Presenter	Topic Covered
0:00 - 3:00 (3 mins)	Israel John	Project Overview & Business Logic: Introduction, Problem Statement, Financial Business Rules (BRD).
3:00 - 8:00 (5 mins)	Samuel Enyi	Complete Database Architecture: ERD Design, 3NF Normalization, DDL Scripts, Constraints, Foreign Keys, SQL Views, & Query Performance Tuning (<code>EXPLAIN ANALYZE</code>).
8:00 - 10:00 (2 mins)	Ruth Thompson	Data Engineering & Integrity: Synthetic Data Generation (15,000+ records), Data Dictionary, and QA Validation.
10:00 - 13:00 (3 mins)	Christian Nwobodo	Financial BI Dashboard Demo: Live presentation of Revenue, Total Deposits, Balances, and Net Margin.
13:00 - 15:00 (2 mins)	Richmond Akara-Nwaogu	Credit Risk BI Dashboard Demo: Live presentation of NPL %, Delinquencies, Credit Scores, and Risk Segmentation.

Architected and Structured by Samuel Enyi, Lead Database Architect & SQL Engineer.

PROJECT CONCORD: LEAD DATA ARCHITECT GUIDE

Samuel Enyi's Step-by-Step SQL Engineering & Database Release Roadmap

This guide defines the precise engineering execution steps for **Samuel Enyi** to build, secure, optimize, and hand over the **Veridian Unified Data Platform** database.

1. Technical Task Breakdown & Sequence



2. Step-by-Step Technical Guide

Step 1: Conceptual & Logical ERD Setup

- **Action:** Use [dbdiagram.io](#).
- **Rule:** Map the **6 Core Services Hub entities** in the center and the **5 divisional modules** around it.
- **Goal:** Define primary keys and foreign key relationships with clear cardinalities before writing any DDL code.

Step 2: Implement Core Services Hub DDL

Create the shared foundation tables. Apply naming standards (snake_case, plural table names, singular `_id` keys):

- **customers** : Contains `customer_id` (UUID or SERIAL), `full_name` , `email` , `consent_retail_loyalty` (Boolean), `consent_vfs_credit_sharing` (Boolean).
- **employees** : Contains `employee_id` , `full_name` , `division_id` , `role_title` , `reports_to` (self-referential FK).
- **suppliers_vendors** : Unified registry. Farmers are flagged as `supplier_type = 'FARMER'` .
- **locations_sites** : Shared geocoordinates (`latitude DECIMAL(9,6)` , `longitude DECIMAL(9,6)`).
- **product_service_catalogue** : Shared SKUs (`product_id` , `product_name` , `unit_of_measure`).
- **financial_account_references** : References VFS account IDs (`account_ref_id` , `customer_id` , `account_type` , `status`) without storing private ledger columns.

Step 3: Implement Divisional Module Tables

Create specific tables for the divisions. Ensure strict constraints to maintain business logic integrity:

- `pos_transactions` : Add `CHECK (total_amount > 0)` .
- `inventory_stock_levels` : Add `CHECK (quantity_on_hand >= 0)` .
- `loans` : Add `CHECK (principal_amount > 0)` .
- `leases` : Add `CHECK (end_date > start_date)` .
- `harvest_batches` : Link back to Core Hub (`farm_id` , `product_id`).

Step 4: Write Row Level Security (RLS) Policies

Enforce security at the database engine level (PostgreSQL) rather than application code:

1. Enable RLS:

```
sql ALTER TABLE veridian_financial_services.wallet_accounts ENABLE ROW LEVEL SECURITY; ALTER TABLE veridian_financial_services.loans
ENABLE ROW LEVEL SECURITY;
```

2. Define Access Roles:

```
sql CREATE ROLE retail_staff; CREATE ROLE financial_staff;
```

3. Write Consent-based Policies:

```
sql -- Allow VFS staff to view AgriCore harvest details ONLY IF the farmer has consented: CREATE POLICY vfs_farmer_view_policy ON
agricore.farms FOR SELECT TO financial_staff USING ( EXISTS ( SELECT 1 FROM core.customers c WHERE c.customer_id = farms.supplier_id
AND c.consent_vfs_credit_sharing = TRUE ) );
```

Step 5: Execute Synthetic Data Ingestion

- Coordinate with **Ruth Thompson** to get the generated CSV data files.
- Use pgAdmin Import/Export Tool or PostgreSQL `COPY` command to seed data.
- Ensure correct loading sequence: **Core Hub tables first**, followed by divisional transactional tables.

Step 6: Create Curated Analytical Views

Expose clean, pre-joined views so the BI developers don't have to write complex joins.

View 1: Ngozi's Store Stock & Supply Alert

```
CREATE VIEW v_agricore_retail_supply_alert AS
SELECT
    s.store_id,
    l.site_name AS store_location,
    p.product_name,
    isl.quantity_on_hand AS retail_stock_level,
    hb.volume_kg AS recent_harvest_volume,
    hb.harvest_date
FROM retail.stores s
JOIN core.locations_sites l ON s.location_id = l.location_id
JOIN retail.inventory_stock_levels isl ON s.store_id = isl.store_id
JOIN core.product_service_catalogue p ON isl.product_id = p.product_id
LEFT JOIN agricore.harvest_batches hb ON p.product_id = hb.product_id;
```

View 2: Chinedu's Farmer Credit Evaluation View

```
CREATE VIEW v_farmer_credit_evaluation AS
SELECT
    c.customer_id,
    c.full_name AS farmer_name,
    f.farm_id,
    f.primary_crop,
    SUM(hb.volume_kg) AS total_harvested_volume_kg,
    c.consent_vfs_credit_sharing
FROM core.customers c
JOIN agricore.farms f ON c.customer_id = f.supplier_id
JOIN agricore.harvest_batches hb ON f.farm_id = hb.farm_id
WHERE c.consent_vfs_credit_sharing = TRUE
GROUP BY c.customer_id, c.full_name, f.farm_id, f.primary_crop;
```

3. Database Connection Parameters (Handover Package)

Once database script is loaded and views are created, copy the connection details below and hand them over to **Christian** and **Richmond** to connect Power BI:

- **Host:** db.supabase.co (replace with your Supabase Host URL)
 - **Port:** 5432
 - **Database:** postgres
 - **Portals / Views to Connect:**
 - v_agricore_retail_supply_alert
 - v_farmer_credit_evaluation
 - v_warehouse_lease_utilization
 - v_executive_consolidated_revenue
-

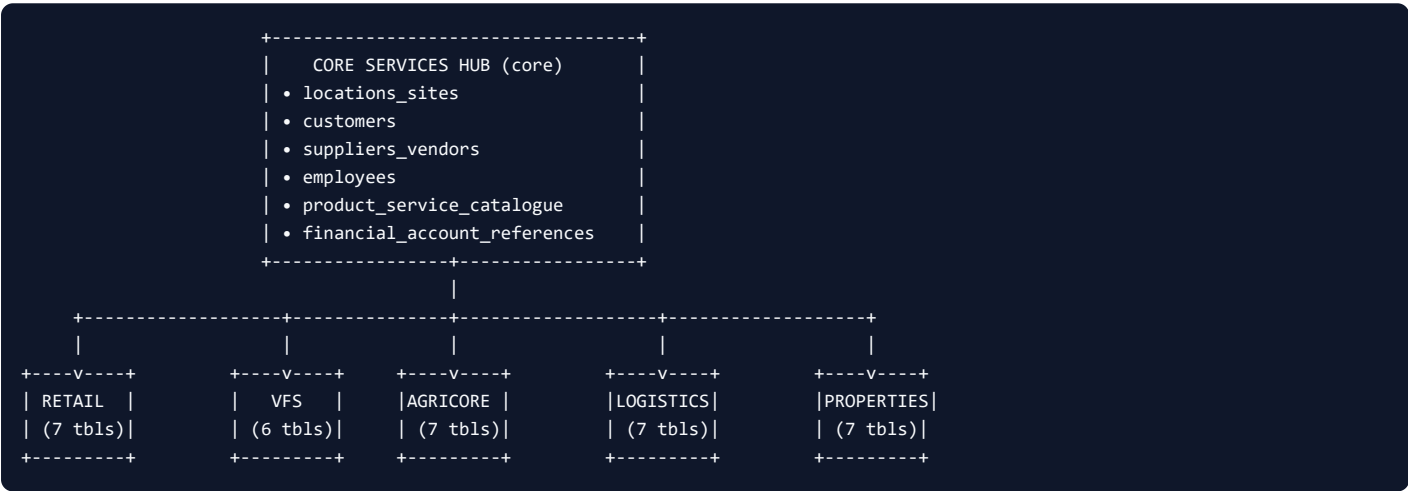
PROJECT CONCORD: VERIDIAN UNIFIED DATA PLATFORM

Master Enterprise Data Dictionary (All 43 Entities)

Document Reference: VCH-PC-2026-DD
Version: 1.0 (Final Enterprise Release)
Author: Samuel Enyi (Lead Database Architect & SQL Engineer)
Database Engine: PostgreSQL 15 (Supabase Managed Engine)
Target Defense Date: Friday, 14 August 2026 @ 9:00 AM Prompt

1. Executive Overview & Schema Organization

The **Veridian Unified Data Platform** follows a **Core Hub & Divisional Modules** architecture. The platform consists of **6 distinct logical schemas** housing **43 primary operational entities** (and associated analytical views):



2. Core Services Hub (core Schema — 6 Shared Entities)

2.1 core.locations_sites

- Purpose:** Single source of truth for all physical and operational sites across Nigeria (NG), Ghana (GH), and Kenya (KE).
- Referenced By:** retail.stores , agricore.farms , agricore.processing_runs , logistics.shipments , logistics.routes , logistics.warehouses , properties.properties .

Column Name	Data Type	Constraints	Description / Business Rules
location_id	VARCHAR(30)	PRIMARY KEY	Unique location identifier (e.g., LOC-NG-LOS-001).
site_name	VARCHAR(150)	NOT NULL	Descriptive name of the site (e.g., Ikeja Hypermarket Premises).
site_type	VARCHAR(50)	NOT NULL , CHECK	Type of facility: 'STORE' , 'WAREHOUSE' , 'FARM' , 'PROPERTY' , 'OFFICE' .
address	TEXT	NOT NULL	Physical street address.
city	VARCHAR(100)	NOT NULL	City or municipality (e.g., Lagos , Kano , Accra).
country_code	VARCHAR(5)	NOT NULL , CHECK	ISO 2-letter country code: 'NG' , 'GH' , 'KE' .
latitude	DECIMAL(9,6)	Nullable	Geolocation latitude coordinate.
longitude	DECIMAL(9,6)	Nullable	Geolocation longitude coordinate.
created_at	TIMESTAMPZ	DEFAULT NOW()	Record creation timestamp.

2.2 core.customers

- **Purpose:** Unified identity master for all individual customers interacting with Meridian Retail and Veridian Financial Services.
- **Referenced By:** retail.pos_transactions , retail.loyalty_accounts , vfs.wallet_accounts , vfs.loans , vfs.kyc_records , core.financial_account_references .

Column Name	Data Type	Constraints	Description / Business Rules
customer_id	VARCHAR(30)	PRIMARY KEY	Global customer ID (e.g., CUST-100452).
full_name	VARCHAR(150)	NOT NULL	Legal full name of the individual customer.
date_of_birth	DATE	NOT NULL	Date of birth for KYC and age verification.
primary_contact	VARCHAR(50)	NOT NULL	Phone number or primary email address.
registered_country	VARCHAR(5)	NOT NULL , CHECK	Registration country: 'NG' , 'GH' , 'KE' .
consent_retail_loyalty	BOOLEAN	DEFAULT TRUE	Consent flag for retail marketing & loyalty program.
consent_vfs_credit_sharing	BOOLEAN	DEFAULT FALSE	Consent flag permitting VFS credit assessment access.
created_date	TIMESTAMP TZ	DEFAULT NOW()	Customer registration timestamp.

2.3 core.suppliers_vendors

- **Purpose:** Shared master registry for external suppliers, vendor partners, and AgriCore smallholder farmers.
- **Referenced By:** agricore.farmers , agricore.farms , retail.supplier_deliveries , vfs.loans .

Column Name	Data Type	Constraints	Description / Business Rules
supplier_id	VARCHAR(30)	PRIMARY KEY	Global supplier identifier (e.g., SUPP-AGRI-0012).
legal_name	VARCHAR(150)	NOT NULL	Legal business name or farmer full name.
supplier_type	VARCHAR(50)	NOT NULL , CHECK	Classification: 'FARMER' , 'WHOLESALE_VENDOR' , 'SERVICE_PROVIDER' .
primary_division_id	VARCHAR(30)	NOT NULL , CHECK	Primary division relationship: 'AGRICORE' , 'RETAIL' , 'LOGISTICS' , 'VFS' , 'PROPERTIES' .
onboarding_date	DATE	NOT NULL	Date supplier was onboarded.
status	VARCHAR(30)	DEFAULT 'ACTIVE' , CHECK	Account status: 'ACTIVE' , 'SUSPENDED' , 'INACTIVE' .

2.4 core.employees

- **Purpose:** Group-wide employee registry across all 5 operating divisions and executive leadership.
- **Referenced By:** retail.stores (manager), agricore.harvest_batches (agent), agricore.quality_grades (inspector), logistics.drivers (driver employee).

Column Name	Data Type	Constraints	Description / Business Rules
employee_id	VARCHAR(30)	PRIMARY KEY	Unique staff ID (e.g., EMP-10023).
full_name	VARCHAR(150)	NOT NULL	Employee full name.
division_id	VARCHAR(30)	NOT NULL , CHECK	Operating division: 'RETAIL' , 'LOGISTICS' , 'AGRICORE' , 'VFS' , 'PROPERTIES' , 'EXECUTIVE' .
role_title	VARCHAR(100)	NOT NULL	Job designation (e.g., Store Manager , Field Agent).
employment_status	VARCHAR(30)	DEFAULT 'ACTIVE' , CHECK	Status: 'ACTIVE' , 'ON_LEAVE' , 'TERMINATED' .
hire_date	DATE	NOT NULL	Official date of employment.
reports_to	VARCHAR(30)	FK -> core.employees	Manager's employee ID (Self-referential hierarchy).

2.5 core.product_service_catalogue

- **Purpose:** Master product and service master list sold through Meridian Retail or produced by AgriCore.
- **Referenced By:** retail.transaction_line_items , retail.inventory_stock_levels , retail.promotions , agricore.harvest_batches .

Column Name	Data Type	Constraints	Description / Business Rules
product_id	VARCHAR(30)	PRIMARY KEY	Unique product SKU code (e.g., PROD-GRN-001).
product_name	VARCHAR(150)	NOT NULL	Commercial name (e.g., Yellow Maize Grain 50kg).
category	VARCHAR(50)	NOT NULL , CHECK	Category: 'FRESH_PRODUCE' , 'GRAINS' , 'PACKAGED_GOODS' , 'BEVERAGES' , 'SERVICES' .
unit_of_measure	VARCHAR(20)	NOT NULL , CHECK	Standard unit: 'KG' , 'TON' , 'BAG' , 'CARTON' , 'LITRE' , 'UNIT' .
primary_division_id	VARCHAR(30)	NOT NULL , CHECK	Originating division: 'AGRICORE' , 'RETAIL' .
is_active	BOOLEAN	DEFAULT TRUE	Active sellable status.

2.6 core.financial_account_references

- **Purpose:** Controlled, non-sensitive reference linking customers and suppliers to VFS financial product records without exposing banking details.
- **Referenced By:** vfs.wallet_accounts .

Column Name	Data Type	Constraints	Description / Business Rules
account_ref_id	VARCHAR(30)	PRIMARY KEY	Reference code (e.g., FAR-WLT-90021).
customer_id	VARCHAR(30)	NOT NULL , FK -> core.customers	Link to customer identity.
account_type	VARCHAR(30)	NOT NULL , CHECK	Product type: 'SAVINGS' , 'WALLET' , 'LOAN' .
account_status	VARCHAR(30)	DEFAULT 'ACTIVE' , CHECK	Status: 'ACTIVE' , 'DORMANT' , 'CLOSED' .
opened_date	DATE	NOT NULL	Account opening date.

3. Meridian Retail & Consumer Module (retail Schema — 7 Entities)

3.1 retail.stores

Column Name	Data Type	Constraints	Description
store_id	VARCHAR(30)	PRIMARY KEY	Unique branch identifier (e.g., STR-LOS-01).
location_id	VARCHAR(30)	NOT NULL , FK -> core.locations_sites	Physical site mapping.
store_format	VARCHAR(50)	NOT NULL , CHECK	'HYPERMARKET' , 'SUPERMARKET' , 'EXPRESS' , 'ONLINE_FULFILLMENT' .
opening_date	DATE	NOT NULL	Branch opening date.
manager_employee_id	VARCHAR(30)	FK -> core.employees	Assigned store manager.
property_id	VARCHAR(30)	FK -> properties.properties	Premises lease mapping.

3.2 retail.pos_transactions

Column Name	Data Type	Constraints	Description
transaction_id	VARCHAR(40)	PRIMARY KEY	Point-of-sale receipt ID.
store_id	VARCHAR(30)	NOT NULL , FK -> retail.stores	Store where transaction occurred.
customer_id	VARCHAR(30)	FK -> core.customers	Transacting customer (if loyalty/identified).
transaction_date	TIMESTAMPZ	NOT NULL , DEFAULT NOW()	Date and time of sale.
total_amount	DECIMAL(18,2)	NOT NULL , CHECK (> 0)	Total bill value in local currency (NGN).
payment_method	VARCHAR(30)	NOT NULL , CHECK	'CASH' , 'VFS_WALLET' , 'DEBIT_CARD' , 'POS' .

3.3 retail.transaction_line_items

Column Name	Data Type	Constraints	Description
line_item_id	VARCHAR(40)	PRIMARY KEY	Unique line item UUID/ID.
transaction_id	VARCHAR(40)	NOT NULL , FK -> retail.pos_transactions	Header transaction reference.
product_id	VARCHAR(30)	NOT NULL , FK -> core.product_service_catalogue	Product purchased.
quantity	INT	NOT NULL , CHECK (> 0)	Units purchased.
unit_price	DECIMAL(18,2)	NOT NULL , CHECK (> 0)	Unit selling price.

3.4 retail.inventory_stock_levels

Column Name	Data Type	Constraints	Description
stock_id	VARCHAR(40)	PRIMARY KEY	Stock record ID.
store_id	VARCHAR(30)	NOT NULL , FK -> retail.stores	Store location.
product_id	VARCHAR(30)	NOT NULL , FK -> core.product_service_catalogue	Product SKU.
quantity_on_hand	INT	NOT NULL , CHECK (>= 0)	Current stock count.
last_counted_date	TIMESTAMPZ	DEFAULT NOW()	Last physical audit date.

3.5 retail.promotions

Column Name	Data Type	Constraints	Description
promotion_id	VARCHAR(30)	PRIMARY KEY	Promo code ID.
product_id	VARCHAR(30)	NOT NULL , FK -> core.product_service_catalogue	Discounted product.
discount_type	VARCHAR(30)	NOT NULL , CHECK	'PERCENTAGE' , 'FIXED_AMOUNT' , 'BUY_1_GET_1' .
discount_value	DECIMAL(18,2)	NOT NULL , CHECK (> 0)	Discount rate/amount.
start_date	DATE	NOT NULL	Promo start date.
end_date	DATE	NOT NULL , CHECK (end_date >= start_date)	Promo end date.

3.6 retail.loyalty_accounts

Column Name	Data Type	Constraints	Description
loyalty_id	VARCHAR(30)	PRIMARY KEY	Loyalty card ID.
customer_id	VARCHAR(30)	NOT NULL , FK -> core.customers	Member customer ID.
tier	VARCHAR(20)	DEFAULT 'BRONZE' , CHECK	'BRONZE' , 'SILVER' , 'GOLD' , 'PLATINUM' .
points_balance	INT	DEFAULT 0 , CHECK (>= 0)	Accumulated reward points.
enrolment_date	DATE	NOT NULL , DEFAULT CURRENT_DATE	Date joined loyalty program.

3.7 retail.supplier_deliveries

Column Name	Data Type	Constraints	Description
delivery_id	VARCHAR(40)	PRIMARY KEY	Inbound delivery manifest ID.
supplier_id	VARCHAR(30)	NOT NULL , FK -> core.suppliers_vendors	Supplier (AgriCore or Wholesale).
store_id	VARCHAR(30)	NOT NULL , FK -> retail.stores	Destination retail store.
expected_date	DATE	NOT NULL	Scheduled delivery date.
received_date	DATE	Nullable	Actual receipt date.
status	VARCHAR(30)	DEFAULT 'SCHEDULED' , CHECK	'SCHEDULED' , 'RECEIVED' , 'DELAYED' , 'SHORTFALL' , 'CANCELLED' .

4. Veridian Financial Services Module (vfs Schema — 6 Entities)

4.1 vfs.wallet_accounts

Column Name	Data Type	Constraints	Description
wallet_id	VARCHAR(30)	PRIMARY KEY	Digital wallet account ID.
customer_id	VARCHAR(30)	NOT NULL , FK -> core.customers	Wallet holder.
account_ref_id	VARCHAR(30)	FK -> core.financial_account_references	Core hub account ref.
balance	DECIMAL(18,2)	DEFAULT 0.00 , CHECK (>= 0)	Current wallet funds (NGN).
status	VARCHAR(30)	DEFAULT 'ACTIVE' , CHECK	'ACTIVE' , 'BLOCKED' , 'SUSPENDED' .

4.2 vfs.wallet_transactions

Column Name	Data Type	Constraints	Description
wallet_txn_id	VARCHAR(40)	PRIMARY KEY	Transaction ledger entry ID.
wallet_id	VARCHAR(30)	NOT NULL , FK -> vfs.wallet_accounts	Source wallet.
counterparty_type	VARCHAR(50)	NOT NULL , CHECK	'MERIDIAN_RETAIL_POS' , 'AGRICORE_PAYOUT' , 'TRANSFER' , 'DEPOSIT' , 'WITHDRAWAL' .
amount	DECIMAL(18,2)	NOT NULL , CHECK (> 0)	Transaction value.
transaction_date	TIMESTAMPZ	DEFAULT NOW()	Execution timestamp.

4.3 vfs.loans

Column Name	Data Type	Constraints	Description
loan_id	VARCHAR(30)	PRIMARY KEY	Unique credit facility ID.
borrower_customer_id	VARCHAR(30)	FK -> core.customers	Borrower (if individual customer).
borrower_supplier_id	VARCHAR(30)	FK -> core.suppliers_vendors	Borrower (if farmer/supplier).
principal_amount	DECIMAL(18,2)	NOT NULL , CHECK (> 0)	Approved principal loan amount.
interest_rate	DECIMAL(5,2)	NOT NULL , CHECK (> 0)	Annual percentage interest rate.
status	VARCHAR(30)	DEFAULT 'ACTIVE' , CHECK	'APPLIED' , 'DISBURSED' , 'ACTIVE' , 'PAID_OFF' , 'DEFAULTED' .

4.4 vfs.loan_repayments

Column Name	Data Type	Constraints	Description
repayment_id	VARCHAR(40)	PRIMARY KEY	Repayment schedule entry ID.
loan_id	VARCHAR(30)	NOT NULL , FK -> vfs.loans	Master loan ID.
due_date	DATE	NOT NULL	Scheduled repayment date.
amount_due	DECIMAL(18,2)	NOT NULL , CHECK (> 0)	Required installment amount.
paid_amount	DECIMAL(18,2)	DEFAULT 0.00 , CHECK (>= 0)	Actual amount paid.
paid_date	DATE	Nullable	Payment receipt date.

4.5 vfs.kyc_records

Column Name	Data Type	Constraints	Description
kyc_id	VARCHAR(30)	PRIMARY KEY	Compliance verification ID.
customer_id	VARCHAR(30)	NOT NULL , FK -> core.customers	Customer verified.
verification_level	VARCHAR(20)	NOT NULL , CHECK	Regulatory tier: 'TIER_1' , 'TIER_2' , 'TIER_3' .
verified_date	DATE	NOT NULL , DEFAULT CURRENT_DATE	Approval date.

4.6 vfs.merchant_settlements

Column Name	Data Type	Constraints	Description
settlement_id	VARCHAR(40)	PRIMARY KEY	Daily/weekly settlement batch ID.
division_id	VARCHAR(30)	NOT NULL , CHECK	Recipient division: 'RETAIL' , 'LOGISTICS' , 'AGRICORE' .
settlement_date	DATE	NOT NULL	Settlement date.
total_amount	DECIMAL(18,2)	NOT NULL , CHECK (> 0)	Total net payout amount.

5. AgriCore Agribusiness Module (`agricore` Schema — 7 Entities)

5.1 `agricore.farms`

Column Name	Data Type	Constraints	Description
<code>farm_id</code>	<code>VARCHAR(30)</code>	PRIMARY KEY	Farm plot ID (e.g., <code>FRM-KAD-001</code>).
<code>supplier_id</code>	<code>VARCHAR(30)</code>	NOT NULL , FK -> <code>core.suppliers_vendors</code>	Registered farmer/owner ID.
<code>location_id</code>	<code>VARCHAR(30)</code>	NOT NULL , FK -> <code>core.locations_sites</code>	Geographical location mapping.
<code>size_hectares</code>	<code>DECIMAL(8,2)</code>	NOT NULL , CHECK (> 0)	Plot area in hectares.
<code>primary_crop</code>	<code>VARCHAR(50)</code>	NOT NULL	Dominant crop (e.g., <code>Maize</code> , <code>Plantain</code> , <code>Cowpea</code>).

5.2 `agricore.farmers`

Column Name	Data Type	Constraints	Description
<code>farmer_id</code>	<code>VARCHAR(30)</code>	PRIMARY KEY	Farmer profile ID.
<code>supplier_id</code>	<code>VARCHAR(30)</code>	NOT NULL , FK -> <code>core.suppliers_vendors</code>	Shared supplier core entity.
<code>registration_date</code>	<code>DATE</code>	NOT NULL	Date registered with AgriCore.
<code>cooperative_name</code>	<code>VARCHAR(150)</code>	Nullable	Farmer cooperative association name.

5.3 `agricore.harvest_batches`

Column Name	Data Type	Constraints	Description
<code>harvest_id</code>	<code>VARCHAR(30)</code>	PRIMARY KEY	Harvest batch collection ID.
<code>farm_id</code>	<code>VARCHAR(30)</code>	NOT NULL , FK -> <code>agricore.farms</code>	Source farm.
<code>product_id</code>	<code>VARCHAR(30)</code>	NOT NULL , FK -> <code>core.product_service_catalogue</code>	Harvested commodity SKU.
<code>harvest_date</code>	<code>DATE</code>	NOT NULL	Collection date.
<code>volume_kg</code>	<code>DECIMAL(12,2)</code>	NOT NULL , CHECK (> 0)	Gross harvested volume (kg).
<code>field_agent_employee_id</code>	<code>VARCHAR(30)</code>	FK -> <code>core.employees</code>	AgriCore field collection agent.

5.4 `agricore.processing_runs`

Column Name	Data Type	Constraints	Description
<code>run_id</code>	<code>VARCHAR(30)</code>	PRIMARY KEY	Facility processing batch ID.
<code>harvest_id</code>	<code>VARCHAR(30)</code>	NOT NULL , FK -> <code>agricore.harvest_batches</code>	Input harvest batch.
<code>facility_location_id</code>	<code>VARCHAR(30)</code>	NOT NULL , FK -> <code>core.locations_sites</code>	Processing plant location.
<code>run_date</code>	<code>DATE</code>	NOT NULL	Processing date.
<code>output_volume_kg</code>	<code>DECIMAL(12,2)</code>	NOT NULL , CHECK (>= 0)	Net processed output (kg).

5.5 agricore.quality_grades

Column Name	Data Type	Constraints	Description
grade_id	VARCHAR(30)	PRIMARY KEY	Quality inspection record ID.
run_id	VARCHAR(30)	NOT NULL , FK -> agricore.processing_runs	Processing run inspected.
grade_level	VARCHAR(20)	NOT NULL , CHECK	Quality grade: 'GRADE_A' , 'GRADE_B' , 'GRADE_C' , 'REJECTED' .
moisture_content	DECIMAL(5,2)	Nullable	Moisture percentage measure.
inspector_employee_id	VARCHAR(30)	FK -> core.employees	QA inspector employee ID.

5.6 agricore.wholesale_shipments

Column Name	Data Type	Constraints	Description
wholesale_id	VARCHAR(30)	PRIMARY KEY	Wholesale dispatch ID.
run_id	VARCHAR(30)	NOT NULL , FK -> agricore.processing_runs	Source processed batch.
destination_type	VARCHAR(30)	NOT NULL , CHECK	'MERIDIAN_RETAIL_STORE' , 'THIRD_PARTY_CLIENT' .
destination_id	VARCHAR(30)	NOT NULL	Target store/client ID.
shipment_id	VARCHAR(40)	FK -> logistics.shipments	Transport shipment link.

5.7 agricore.farmer_loans_reference

Column Name	Data Type	Constraints	Description
reference_id	VARCHAR(30)	PRIMARY KEY	Cross-divisional reference ID.
farmer_id	VARCHAR(30)	NOT NULL , FK -> agricore.farmers	AgriCore farmer ID.
loan_id	VARCHAR(30)	NOT NULL , FK -> vfs.loans	VFS loan ID.
visible_summary_status	VARCHAR(50)	NOT NULL	Consent-filtered loan summary status.

6. Concord Logistics Module (logistics Schema — 7 Entities)

6.1 logistics.vehicles

Column Name	Data Type	Constraints	Description
vehicle_id	VARCHAR(30)	PRIMARY KEY	Fleet vehicle ID (e.g., VEH-TRK-012).
registration_number	VARCHAR(30)	UNIQUE , NOT NULL	Official license plate number.
vehicle_type	VARCHAR(50)	NOT NULL , CHECK	'LAST_MILE_VAN' , 'HAULAGE_TRUCK' , 'REFRIGERATED_TRUCK' .
capacity_kg	INT	NOT NULL , CHECK (> 0)	Payload weight limit (kg).
status	VARCHAR(30)	DEFAULT 'AVAILABLE' , CHECK	Fleet status: 'AVAILABLE' , 'IN_TRANSIT' , 'MAINTENANCE' .

6.2 logistics.drivers

Column Name	Data Type	Constraints	Description
driver_id	VARCHAR(30)	PRIMARY KEY	Driver profile ID.
employee_id	VARCHAR(30)	NOT NULL , FK -> core.employees	Driver employee record.
licence_number	VARCHAR(50)	UNIQUE , NOT NULL	Commercial driver license number.
licence_expiry	DATE	NOT NULL	Driver license expiration date.

6.3 logistics.shipments

Column Name	Data Type	Constraints	Description
shipment_id	VARCHAR(40)	PRIMARY KEY	Master freight shipment ID.
origin_location_id	VARCHAR(30)	NOT NULL , FK -> core.locations_sites	Pickup location.
destination_location_id	VARCHAR(30)	NOT NULL , FK -> core.locations_sites	Drop-off location.
client_type	VARCHAR(50)	NOT NULL , CHECK	'INTERNAL_AGRICORE' , 'INTERNAL_RETAIL' , 'THIRD_PARTY_COMMERCIAL' .
status	VARCHAR(30)	DEFAULT 'SCHEDULED' , CHECK	'SCHEDULED' , 'IN_TRANSIT' , 'DELIVERED' , 'DELAYED' , 'CANCELLED' .

6.4 logistics.shipment_legs

Column Name	Data Type	Constraints	Description
leg_id	VARCHAR(40)	PRIMARY KEY	Multi-leg route segment ID.
shipment_id	VARCHAR(40)	NOT NULL , FK -> logistics.shipments	Master shipment ID.
vehicle_id	VARCHAR(30)	NOT NULL , FK -> logistics.vehicles	Assigned vehicle.
driver_id	VARCHAR(30)	NOT NULL , FK -> logistics.drivers	Assigned driver.
departure_time	TIMESTAMPZ	Nullable	Actual departure timestamp.
arrival_time	TIMESTAMPZ	Nullable	Actual arrival timestamp.

6.5 logistics.routes

Column Name	Data Type	Constraints	Description
route_id	VARCHAR(30)	PRIMARY KEY	Standard transit corridor ID.
route_name	VARCHAR(150)	NOT NULL	Corridor name (e.g., Kano to Lagos Haulage).
origin_location_id	VARCHAR(30)	NOT NULL , FK -> core.locations_sites	Origin site.
destination_location_id	VARCHAR(30)	NOT NULL , FK -> core.locations_sites	Destination site.
distance_km	DECIMAL(8,2)	NOT NULL , CHECK (> 0)	Route distance in kilometers.

6.6 logistics.warehouses

Column Name	Data Type	Constraints	Description
warehouse_id	VARCHAR(30)	PRIMARY KEY	Logistics depot ID.
location_id	VARCHAR(30)	NOT NULL , FK -> core.locations_sites	Physical location ID.
capacity_units	INT	NOT NULL , CHECK (> 0)	Pallet/unit storage capacity.
leased_from_property_id	VARCHAR(30)	FK -> properties.properties	Veridian Properties lease reference.

6.7 logistics.maintenance_logs

Column Name	Data Type	Constraints	Description
maintenance_id	VARCHAR(40)	PRIMARY KEY	Fleet repair log entry ID.
vehicle_id	VARCHAR(30)	NOT NULL , FK -> logistics.vehicles	Serviced vehicle.
service_date	DATE	NOT NULL	Maintenance date.
cost	DECIMAL(18,2)	NOT NULL , CHECK (>= 0)	Repair cost in NGN.
description	TEXT	Nullable	Maintenance work summary.

7. Veridian Properties Module (properties Schema — 7 Entities)

7.1 properties.properties

Column Name	Data Type	Constraints	Description
property_id	VARCHAR(30)	PRIMARY KEY	Real estate asset ID (e.g., PROP-LOS-004).
location_id	VARCHAR(30)	NOT NULL , FK -> core.locations_sites	Site location mapping.
property_type	VARCHAR(50)	NOT NULL , CHECK	'RETAIL_STORE_PREMISES' , 'LOGISTICS_WAREHOUSE' , 'AGRI_FACILITY' , 'COMMERCIAL_OFFICE' .
size_sqm	DECIMAL(10,2)	NOT NULL , CHECK (> 0)	Total floor area in square meters.
ownership_status	VARCHAR(30)	NOT NULL , CHECK	'OWNED_FREEHOLD' , 'LEASED_FROM_LANDLORD' .

7.2 properties.tenants

Column Name	Data Type	Constraints	Description
tenant_id	VARCHAR(30)	PRIMARY KEY	Occupant tenant ID.
tenant_type	VARCHAR(30)	NOT NULL , CHECK	'INTERNAL_DIVISION' , 'EXTERNAL_COMMERCIAL_TENANT' .
division_id	VARCHAR(30)	Nullable, CHECK	Internal division: 'RETAIL' , 'LOGISTICS' , 'AGRICORE' , 'VFS' .
external_tenant_name	VARCHAR(150)	Nullable	Commercial client company name.

7.3 properties.leases

Column Name	Data Type	Constraints	Description
lease_id	VARCHAR(30)	PRIMARY KEY	Property lease contract ID.
property_id	VARCHAR(30)	NOT NULL , FK -> properties.properties	Leased property asset.
tenant_id	VARCHAR(30)	NOT NULL , FK -> properties.tenants	Tenant contract holder.
start_date	DATE	NOT NULL	Lease commencement date.
end_date	DATE	NOT NULL	Lease expiration date (CHECK end_date > start_date).
monthly_rent	DECIMAL(18,2)	NOT NULL , CHECK (> 0)	Monthly rental fee in NGN.

7.4 properties.maintenance_requests

Column Name	Data Type	Constraints	Description
request_id	VARCHAR(40)	PRIMARY KEY	Property defect/repair request ID.
property_id	VARCHAR(30)	NOT NULL , FK -> properties.properties	Target property asset.
requested_date	DATE	NOT NULL , DEFAULT CURRENT_DATE	Date ticket opened.
category	VARCHAR(50)	NOT NULL	Defect category (e.g., Roofing , HVAC , Electrical).
status	VARCHAR(30)	DEFAULT 'OPEN' , CHECK	'OPEN' , 'IN_PROGRESS' , 'RESOLVED' .
resolved_date	DATE	Nullable	Ticket resolution date.

7.5 properties.property_valuations

Column Name	Data Type	Constraints	Description
valuation_id	VARCHAR(40)	PRIMARY KEY	Portfolio audit valuation ID.
property_id	VARCHAR(30)	NOT NULL , FK -> properties.properties	Valued asset ID.
valuation_date	DATE	NOT NULL	Valuation audit date.
assessed_value	DECIMAL(18,2)	NOT NULL , CHECK (> 0)	Assessed market value in NGN.

7.6 properties.utility_accounts

Column Name	Data Type	Constraints	Description
utility_id	VARCHAR(30)	PRIMARY KEY	Utility account mapping ID.
property_id	VARCHAR(30)	NOT NULL , FK -> properties.properties	Property premises.
utility_type	VARCHAR(30)	NOT NULL , CHECK	'ELECTRICITY' , 'WATER' , 'WASTE_MANAGEMENT' .
provider_name	VARCHAR(100)	NOT NULL	Utility provider name (e.g., IKEDC , LWC).
account_number	VARCHAR(50)	NOT NULL	Meter / account number.

7.7 properties.facility_assets

Column Name	Data Type	Constraints	Description
asset_id	VARCHAR(40)	PRIMARY KEY	Capital equipment ID.
property_id	VARCHAR(30)	NOT NULL , FK -> properties.properties	Property location.
asset_type	VARCHAR(50)	NOT NULL	Asset classification (e.g., Generator 500kVA , Chiller Unit).
installed_date	DATE	NOT NULL	Installation date.
condition_rating	VARCHAR(30)	CHECK	Condition: 'EXCELLENT' , 'GOOD' , 'FAIR' , 'NEEDS_REPLACEMENT' .

8. Summary of Key Analytical Views Exposed for BI

View Name	Schema	Base Tables Joined	Key Purpose / Scenario Addressed
v_agricore_retail_supply_alert	public	retail.supplier_deliveries , agricore.harvest_batches , core.product_service_catalogue , retail.stores	Scenario 1 (Ngozi): Provides store managers advance supply shortfall warnings.
v_farmer_credit_assessment	public	agricore.farmers , agricore.harvest_batches , vfs.loans , core.customers , core.suppliers_vendors	Scenario 2 (Chinedu): Consent-based multi-year farmer harvest history for credit scoring.
v_logistics_warehouse_leases	public	logistics.warehouses , properties.properties , properties.leases , properties.tenants	Scenario 3 (Funmi): Real-time warehouse lease status and double-billing prevention.
v_consolidated_group_revenue	public	retail.pos_transactions , logistics.shipments , vfs.wallet_transactions , agricore.wholesale_shipments , properties.leases	Scenario 4 (Adaeze): Same-day cross-divisional revenue consolidation for executive oversight.

Architected and Structured by Samuel Enyi, Lead Database Architect & SQL Engineer.

Project Concord: Power BI Dashboard Build Guide

Ref: VCH-PC-2026-POWERBI-GUIDE | **Version:** 3.0 (Unified Design System)
Author: Samuel Enyi | **Defense Date:** 20 August 2026
Developers: Christian Nwobodo & Richmond Akara-Nwaogu

PART A: UNDERSTAND THE PROBLEM FIRST

Read this before touching Power BI. Your dashboards must answer specific business questions. If you do not know the questions, you cannot build the right answers.

The 4 Business Problems Your Dashboards Solve

#	Problem	Who It Hurts	What Goes Wrong
1	21-Day Revenue Delay	CEO Adaeze	After month-end, all 5 divisions email separate spreadsheets. Finance manually combines them. It takes 21 days. 15.3 billion naira wasted yearly.
2	Invisible Farmer	VFS Loan Officer Chinedu	A farmer with 6 years of reliable AgriCore deliveries applies for a loan. Chinedu only sees 2 years of banking history. VFS and AgriCore never shared data.
3	Double-Booked Warehouse	Logistics Dispatcher Funmi	Funmi sends a truck to a depot bay. When it arrives, the bay is already leased to someone else. Properties changed it by email and Funmi never saw it.
4	Empty Shelf	Store Manager Ngozi	AgriCore farms lose 40% of their harvest to bad weather. Nobody tells Ngozi. Her Enugu store runs out of stock 4 days later with no warning.

How the Database Fixed This

Samuel built 40 tables across 6 database schemas (`core` , `retail` , `vfs` , `agricore` , `logistics` , `properties`). All divisions now share **6 common master tables** in the `core` schema:

Shared Table	What It Connects
<code>core.customers</code>	Same customer ID used by both Retail and VFS
<code>core.suppliers_vendors</code>	Same supplier ID used by both AgriCore and Retail
<code>core.employees</code>	All staff across all 5 divisions
<code>core.locations_sites</code>	All stores, warehouses, and farms share one address system
<code>core.product_service_catalogue</code>	Produce harvested by AgriCore = products sold in Meridian Retail
<code>core.financial_account_references</code>	Non-sensitive bridge so other divisions can see VFS account existence

These cross-division queries are pre-built as **4 SQL Views** in the database:

SQL View	Problem It Solves
<code>core.v_executive_consolidated_revenue</code>	Adaeze: Group revenue in 3 seconds, not 21 days
<code>vfs.v_farmer_credit_evaluation</code>	Chinedu: See 6 years of a farmer harvest history
<code>logistics.v_warehouse_lease_utilization</code>	Funmi: Live lease expiry status per depot
<code>retail.v_agricore_retail_supply_alert</code>	Ngozi: Harvest shortfall alert 4 days in advance

Where Your Dashboard Fits

- Database + Views** = solved the data problem (numbers are now connected across all divisions)
- Your Power BI Dashboard** = solves the human problem (makes the data readable for business users)

The dashboards are the final layer. Without them, only engineers can access this platform.

PART B: UNIFIED VCH CORPORATE COLOR SYSTEM (BOTH DEVELOPERS MUST FOLLOW)

To keep the dashboards clean, consistent, and professional, **both Christian and Richmond will use the exact same corporate color system across all 4 pages.**

Avoid using too many random colors. Keep standard charts neutral, and only use alert colors (Red/Amber/Green) when highlighting specific operational statuses or warnings.

Master Palette Table

Element	Color Name	Hex Code	Usage
Page Background	Corporate Slate	#0F172A	Background for ALL dashboard pages
Card / Container Background	Dark Slate Container	#1E293B	Background for ALL cards, visual containers, and tables (high contrast)
Primary Accent & Headers	Sky Cyan	#38BDF8	Titles, primary bar chart bars, line chart active series, card borders
Primary Text	Crisp White	#F8FAFC	All main text, numbers, and KPI values (100% readable)
Secondary Text	Muted Grey	#94A3B8	Subtitles, axis labels, table column headers
Status: Good / Healthy / Active	Emerald Green	#22C55E	Active leases, healthy stock (>200 units), paid loans, NDPA consent TRUE
Status: Warning / Expiring / Low	Amber Orange	#F59E0B	Low stock (50-200 units), lease expiring in 90 days, active loan
Status: Critical / Expired / Default	Coral Red	#EF4444	Stock critical (<50 units), expired lease, defaulted loan, delayed shipment

PART C: CONNECT TO THE DATABASE (BOTH DEVELOPERS DO THIS)

Step 1 — Install the Driver

- 1. Go to <https://www.npgsql.org/> and download the Npgsql PostgreSQL driver
- 2. Install it, then restart Power BI Desktop

Step 2 — Open a New Connection

- 1. Open **Power BI Desktop**
- 2. Click **Home** ribbon → **Get Data**
- 3. Search for `PostgreSQL` → select **PostgreSQL database** → click **Connect**

Step 3 — Enter Server Details

- **Server:** `aws-1-eu-west-2.pooler.supabase.com:6543`
- **Database:** `postgres`
- **Data Connectivity mode:** `Import`

Click **OK**.

Step 4 — Enter Login Credentials

When the login window appears, click the **Database** tab (not Windows):

- **User name:** `postgres.hraipqaksitkchkhdye`
- **Password:** `Fi4nMf2MXD668s3x`

Click **Connect**.

Step 5 — Select Your Data

- 1. The **Navigator panel** opens showing all schemas
- 2. Expand the schema folder (e.g., `core` , `retail` , `vfs`)
- 3. Tick the checkbox next to each view or table you need
- 4. Click **Load** at the bottom right

PART D: CHRISTIAN NWOBODO'S DASHBOARD

Dashboard File Name: VCH_Executive_Retail_Intelligence_Dashboard.pbix

Your job — answer 2 questions:

- Page 1: What is VCH group revenue across all divisions right now? (Adaeze)
- Page 2: Which stores are about to run out of stock? (Ngozi)

D1 — Load These Data Sources

Primary Views:

What to Load	Where in Navigator	Key Columns
core.v_executive_consolidated_revenue	Expand core then tick the view	division , total_revenue_ngn , transaction_count , last_transaction_date
retail.v_agricore_retail_supply_alert	Expand retail then tick the view	store_city , product_name , retail_stock_on_hand , recent_harvest_volume_kg , delivery_status

Supporting Tables:

Table	Schema	Used For
pos_transactions	retail	Monthly sales trend chart
stores	retail	Store format filter
inventory_stock_levels	retail	Stock quantity visuals
product_service_catalogue	core	Product category filter
locations_sites	core	City and country filter

D2 — Build Page 1: Executive Revenue Command Centre

Style: Professional executive cockpit. Dark Slate background with high-contrast card containers.

Apply Colors from Master Palette (Part B):

- Page background: #0F172A
- Card background: #1E293B
- Main Accent: #38BDF8
- Text: #F8FAFC (Primary), #94A3B8 (Secondary)

Build top to bottom in 5 rows:

ROW 1 — Header

1. Insert → Text Box → stretch full width at top
2. Title: VCH Group Executive Performance Dashboard (Color: #38BDF8 , Bold)
3. Subtitle: Live Cross-Divisional Revenue Intelligence | Powered by Project Concord (Color: #94A3B8)
4. Right Text Box: Last Refreshed: [today's date]

ROW 2 — 4 KPI Cards (equal width, side by side)

For each card: Card Visual → set Background Color to #1E293B .

Card	Field	Format
Total Group Revenue	total_revenue_ngn (SUM) from revenue view	Billions with Naira symbol (₦)
Total Transactions	transaction_count (SUM) from revenue view	Number with comma separators
Top Division	Create DAX measure (see D4)	Text string
Lowest Division	Create DAX measure (see D4)	Text string

ROW 3 — 2 Charts (side by side)

LEFT (60% of row width) — Revenue by Division

- Visual: **Clustered Bar Chart**
- X-axis: `division` from `v_executive_consolidated_revenue`
- Y-axis: `total_revenue_ngn` (SUM)
- Bar color: `#38BDF8` (Sky Cyan for consistency)
- Sort bars descending (highest revenue on top)
- Data Labels: ON (Color: `#F8FAFC`)

RIGHT (40% of row width) — Transactions by Division

- Visual: **Donut Chart**
- Values: `transaction_count`
- Legend: `division`
- Colors: Monochromatic blues/cyans (`#38BDF8` , `#0284C7` , `#0369A1` , `#075985`)

ROW 4 — Monthly Sales Trend (full width)

- Visual: **Line Chart**
- X-axis: `transaction_date` from `pos_transactions` → group by **Month**
- Y-axis: SUM of `total_amount`
- Legend: `payment_method`
- Analytics pane: add an **Average Line** (`#94A3B8` dashed)
- Title: `Meridian Retail Monthly Sales Trend by Payment Method`

ROW 5 — Summary Table (full width)

- Visual: **Table**
- Source: `v_executive_consolidated_revenue`
- Columns: `division` , `total_revenue_ngn` , `transaction_count` , `last_transaction_date`
- Table background: `#1E293B` , Header font: `#38BDF8`
- Conditional formatting on `total_revenue_ngn` :
- Top value → `#22C55E` (Green)
- Middle values → `#38BDF8` (Cyan)
- Lowest value → `#EF4444` (Red)

D3 — Build Page 2: Retail Supply Alert System

Style: Operational alert view. Same slate background, traffic-light status colors for stock alerts only.

Build top to bottom in 5 rows:

ROW 1 — Header

- Title: `Meridian Retail – AgriCore Supply Chain Alert System` (`#38BDF8`)
- Subtitle: `Real-Time Harvest-to-Shelf Supply Monitoring` (`#94A3B8`)

ROW 2 — 3 KPI Cards

Card	How to Build
Total Stores Monitored	Card → <code>store_id</code> from supply alert view → Count Distinct
Products with Stock Risk	Use DAX measure <code>Stock Risk Count</code> (see D4) — Card BG <code>#1E293B</code> , Number color <code>#EF4444</code>
Active Deliveries	Card → <code>delivery_status</code> → count excluding <code>NO_DELIVERY_LOGGED</code>

ROW 3 — Alert Status Table (full width)

- Visual: **Table**
- Source: `retail.v_agricore_retail_supply_alert`
- Columns: `store_city` , `product_name` , `retail_stock_on_hand` , `recent_harvest_volume_kg` , `delivery_status`
- Conditional formatting on `retail_stock_on_hand` (Background color):
- Below 50 → Red `#EF4444` (CRITICAL)
- 50 to 200 → Amber `#F59E0B` (LOW)
- Above 200 → Green `#22C55E` (HEALTHY)

ROW 4 — 2 Charts

LEFT — Stock by Store

- Visual: **Horizontal Bar Chart**
- Y-axis: `store_city`

- X-axis: retail_stock_on_hand
- Sort ascending (lowest stock at top)
- Bar color: #38BDF8

RIGHT — Harvest Volume by Product

- Visual: **Treemap**
- Category: product_name
- Values: recent_harvest_volume_kg
- Color gradient: Slate blue to Sky Cyan (#1E3A5F to #38BDF8)

ROW 5 — Slicers

- 3 slicers along bottom: store_city , product_name , delivery_status
- Slicer container background: #1E293B

D4 — DAX Measures for Christian

```
Total Group Revenue = SUM(v_executive_consolidated_revenue[total_revenue_ngn])

Total Transactions = SUM(v_executive_consolidated_revenue[transaction_count])

Stock Risk Count =
    CALCULATE(
        COUNTROWS(v_agricore_retail_supply_alert),
        v_agricore_retail_supply_alert[retail_stock_on_hand] < 100
    )
```

PART E: RICHMOND AKARA-NWAOGU'S DASHBOARD

Dashboard File Name: VCH_Credit_Risk_Depot_Utilization_Dashboard.pbix

Your job — answer 2 questions:

- Page 1: Is this farmer creditworthy based on their full harvest history? (Chinedu)
- Page 2: Which depots have expired or expiring leases? (Funmi)

E1 — Load These Data Sources

Primary Views:

What to Load	Where in Navigator	Key Columns
vfs.v_farmer_credit_evaluation	Expand vfs then tick the view	farmer_name , farm_id , primary_crop , total_harvest_batches , cumulative_harvest_volume_kg , first_harvest_date , last_harvest_date , consent_vfs_credit_sharing
logistics.v_warehouse_lease_utilization	Expand logistics then tick the view	depot_name , city , total_pallet_capacity , ownership_status , monthly_rent , lease_start , lease_end , lease_status

Supporting Tables:

Table	Schema	Used For
loans	vfs	Loan status and value KPI cards
loan_repayments	vfs	Repayment trend chart
harvest_batches	agricore	Harvest trend over time
warehouses	logistics	Depot capacity bar chart
shipments	logistics	Shipment status chart
properties	properties	Property ownership pie chart

E2 — Build Page 1: Farmer Credit Risk Intelligence Panel

Style: Credit bureau panel. Exactly same slate background (#0F172A) and container background (#1E293B) as Christian's dashboard.

Build top to bottom in 6 rows:

ROW 1 — Header

- Title: VFS Microfinance – Farmer Credit Risk Intelligence Panel (#38BDF8)
- Subtitle: AgriCore-Integrated Cross-Divisional Credit Evaluation System (#94A3B8)

ROW 2 — 4 KPI Cards (Background: #1E293B)

Card	How to Build
Total Farmers Evaluated	Use DAX measure Total Farmers (see E4)
Total Active Loans	status from loans filtered to ACTIVE → Count
Total Loan Portfolio	SUM of principal_amount from loans
Avg Harvest Per Farmer	Use DAX measure Avg Harvest Per Farmer (see E4)

ROW 3 — 2 Charts

LEFT — Top 10 Farmers by Harvest Volume

- Visual: **Horizontal Bar Chart**
- Y-axis: farmer_name from v_farmer_credit_evaluation
- X-axis: cumulative_harvest_volume_kg
- Bar color: #38BDF8 (Sky Cyan)
- Top N filter: Top 10 by volume
- Title: Top 10 Farmers – 6-Year Cumulative Supply Volume

RIGHT — Harvest Mix by Crop

- Visual: **Donut Chart**
- Values: cumulative_harvest_volume_kg
- Legend: primary_crop
- Color palette: Monochromatic blues/cyans (#38BDF8 , #0284C7 , #0369A1 , #075985)

ROW 4 — Harvest Trend (full width)

- Visual: **Line Chart**
- Source: agricore.harvest_batches
- X-axis: harvest_date → group by Month
- Y-axis: SUM of volume_kg
- Legend: primary_crop (top 5 crops)
- Title: AgriCore Monthly Harvest Volume Trend

ROW 5 — Farmer Credit Scorecard Table (full width)

- Visual: **Table**
- Source: vfs.v_farmer_credit_evaluation
- Columns: farmer_name , farm_id , primary_crop , total_harvest_batches , cumulative_harvest_volume_kg , first_harvest_date , last_harvest_date , consent_vfs_credit_sharing
- Sort by cumulative_harvest_volume_kg descending
- Conditional format on consent_vfs_credit_sharing :
 - TRUE → Green cell (#22C55E) — Consent Granted
 - FALSE → Muted grey cell (#64748B) — Consent Restricted (NDPA)

ROW 6 — 2 Charts

LEFT — Loan Status Breakdown

- Visual: **Clustered Bar Chart**
- Source: vfs.loans
- X-axis: status
- Y-axis: Count of loan_id
- Bar colors: PAID_OFF = #22C55E (Green), ACTIVE = #F59E0B (Amber), DEFAULTED = #EF4444 (Red)

RIGHT — Repayment Performance

- Visual: **Stacked Bar Chart**
- Source: vfs.loan_repayments

- X-axis: `due_date` → group by Month
- Y-axis stacks: `amount_due` (`#64748B` grey) then `paid_amount` (`#22C55E` green) on top

E3 — Build Page 2: Warehouse Lease & Depot Utilization Panel

Style: Depot operations panel. Same slate background (`#0F172A`) and containers (`#1E293B`).

Build top to bottom in 6 rows:

ROW 1 — Header

- Title: `Concord Logistics – Warehouse Lease & Depot Capacity Intelligence` (`#38BDF8`)
- Subtitle: `Real-Time Property Lease Status & Bay Availability Monitoring` (`#94A3B8`)

ROW 2 — 4 KPI Cards

Card	How to Build	Number Color
Total Depots	COUNTROWS of <code>v_warehouse_lease_utilization</code>	<code>#F8FAFC</code> (White)
Active Leases	Use DAX <code>Active Leases</code> (see E4)	<code>#22C55E</code> (Green)
Expiring in 90 Days	Use DAX <code>Expiring Leases</code> (see E4)	<code>#F59E0B</code> (Amber)
Expired Leases	Filter <code>lease_status = EXPIRED_LEASE</code> then Count	<code>#EF4444</code> (Red)

ROW 3 — Depot Lease Status Table (full width)

- Visual: **Table**
- Source: `logistics.v_warehouse_lease_utilization`
- Columns: `depot_name` , `city` , `total_pallet_capacity` , `ownership_status` , `monthly_rent` , `lease_start` , `lease_end` , `lease_status`
- Conditional format on `lease_status` (Background color):
 - `ACTIVE_LEASE` → Green `#22C55E`
 - `RENEWAL_DUE_SOON` → Amber `#F59E0B`
 - `EXPIRED_LEASE` → Red `#EF4444`

ROW 4 — 2 Charts

LEFT — Lease Status Distribution

- Visual: **Donut Chart**
- Values: Count of warehouses
- Legend: `lease_status`
- Segment colors: `ACTIVE` = `#22C55E` , `RENEWAL_DUE_SOON` = `#F59E0B` , `EXPIRED` = `#EF4444`

RIGHT — Monthly Rent by City

- Visual: **Bar Chart**
- X-axis: `city`
- Y-axis: SUM of `monthly_rent`
- Bar color: `#38BDF8`

ROW 5 — Depot Capacity (full width)

- Visual: **Horizontal Bar Chart**
- Source: `logistics.warehouses`
- Y-axis: `warehouse_id`
- X-axis: `capacity_units`
- Bar color: `#38BDF8`

ROW 6 — 2 Charts

LEFT — Shipment Status

- Visual: **Clustered Bar Chart**
- Source: `logistics.shipments`
- X-axis: `status`
- Y-axis: Count of shipments
- Bar colors: `DELIVERED` = `#22C55E` , `IN_TRANSIT` = `#F59E0B` , `DELAYED/CANCELLED` = `#EF4444`

RIGHT — Property Ownership Mix

- Visual: **Pie Chart**
- Source: `properties.properties`
- Values: Count of properties

- Legend: `ownership_status`
- Colors: `#38BDF8` (Owned) and `#0284C7` (Leased)

E4 — DAX Measures for Richmond

```
Total Farmers = COUNTROWS(v_farmer_credit_evaluation)

Avg Harvest Per Farmer = AVERAGE(v_farmer_credit_evaluation[cumulative_harvest_volume_kg])

Active Loans = CALCULATE(COUNTROWS(loans), loans[status] = "ACTIVE")

Defaulted Loans = CALCULATE(COUNTROWS(loans), loans[status] = "DEFAULTED")

NPL Ratio = DIVIDE([Defaulted Loans], [Active Loans] + [Defaulted Loans], 0)

Total Loan Portfolio = SUM(loans[principal_amount])

Repayment Rate = DIVIDE(SUM(loan_repayments[paid_amount]), SUM(loan_repayments[amount_due]), 0)

Active Leases =
    CALCULATE(
        COUNTROWS(v_warehouse_lease_utilization),
        v_warehouse_lease_utilization[lease_status] = "ACTIVE_LEASE"
    )

Expiring Leases =
    CALCULATE(
        COUNTROWS(v_warehouse_lease_utilization),
        v_warehouse_lease_utilization[lease_status] = "RENEWAL_DUE_SOON"
    )
```

PART F: FINAL REQUIREMENTS (BOTH DEVELOPERS)

Add Slicers to Every Page

- 1. **Date Range Slicer:** Relative → Last 30 Days / Last 90 Days / Year to Date / All Time
- 2. **Country Slicer:** `country_code` from `core.locations_sites` → filter by NG, GH, KE
- 3. **Division Slicer** (*Christian only, executive page only*)

Slicer background: `#1E293B` , Header text: `#38BDF8` .

Create 3 Bookmarks Per Report (View > Bookmarks > Add Bookmark)

Bookmark Name	Before Saving, Hide These
Executive Summary	All charts and tables — show KPI cards only
Full Detail View	Nothing — show everything
Alert Mode	Filter to show only RED and AMBER rows

Add Page Navigation Buttons

Insert → Buttons → Page Navigator → place at top/bottom of every page.

PART G: DEFENSE PRESENTATION

Before You Start Presenting

- 1. Save your file (File → Save As)
- 2. Press **F11** to go full-screen
- 3. Go to **View** → **Page View** → **Fit to Page**
- 4. Disable Auto Refresh (Home → uncheck scheduled refresh)

What to Say for Each Page

Christian — Page 1 (Executive Revenue):

"This page answers CEO Adaeze's question: what is VCH's total group revenue right now?
Before Project Concord, that answer took 21 days of manual spreadsheet work.
This dashboard pulls from 5 live divisions in under 3 seconds."

Christian — Page 2 (Supply Alert):

"This page tells Ngozi which stores are at risk of running out of stock
before it actually happens, using AgriCore's live harvest data.
Before Project Concord, Retail and AgriCore shared zero data."

Richmond — Page 1 (Farmer Credit):

"This page gives Loan Officer Chinedu a 6-year AgriCore harvest history for any farmer —
a data source that simply did not exist for VFS before Project Concord.
Data only shows for farmers who have given NDPA consent, enforced at the database level."

Richmond — Page 2 (Depot Lease):

"This page shows Funmi which depots have expired leases and which are due for renewal.
Red means expired. Amber means expiring within 90 days.
The double-booking problem is now impossible to miss."

PART H: COMPLETION CHECKLISTS

Christian — tick as you complete:

- [] Connected to Supabase in Power BI Desktop
- [] Loaded `core.v_executive_consolidated_revenue`
- [] Loaded `retail.v_agricore_retail_supply_alert`
- [] Loaded `retail.pos_transactions`, `retail.inventory_stock_levels`, `core.locations_sites`
- [] Applied Unified Palette: Background `#0F172A`, Cards `#1E293B`, Accent `#38BDF8`
- [] Created 3 DAX measures (Total Group Revenue, Total Transactions, Stock Risk Count)
- [] Built Page 1 — Executive Revenue Command Centre (5 rows complete)
- [] Built Page 2 — Retail Supply Alert System (5 rows complete)
- [] Added Date, Country, and Division slicers on all pages
- [] Created 3 bookmarks and page navigation buttons
- [] Saved as `VCH_Executive_Retail_Intelligence_Dashboard.pbix`

Richmond — tick as you complete:

- [] Connected to Supabase in Power BI Desktop
- [] Loaded `vfs.v_farmer_credit_evaluation`
- [] Loaded `logistics.v_warehouse_lease_utilization`
- [] Loaded `vfs.loans`, `vfs.loan_repayments`, `agricore.harvest_batches`
- [] Loaded `logistics.warehouses`, `logistics.shipments`, `properties.properties`
- [] Applied Unified Palette: Background `#0F172A`, Cards `#1E293B`, Accent `#38BDF8`
- [] Created all 9 DAX measures
- [] Built Page 1 — Farmer Credit Risk Intelligence Panel (6 rows complete)
- [] Built Page 2 — Warehouse Lease & Depot Utilization Panel (6 rows complete)
- [] Added Date and Country slicers on all pages
- [] Created 3 bookmarks and page navigation buttons
- [] Saved as `VCH_Credit_Risk_Depot_Utilization_Dashboard.pbix`

Document v3.0 | Samuel Enyi — Lead Database Architect & Technical Director | Project Concord | July 2026

10. Conclusion & Final Governance Recommendations

Project Concord successfully resolves the 20-year data fragmentation across Veridian Consolidated Holdings by implementing a robust, well-governed operational Core Hub surrounded by 5 divisional modules on PostgreSQL (Supabase), connected via an automated ELT pipeline to Google BigQuery, and surfaced through 4 interactive Power BI dashboards.

Core System Accomplishments:

1. **Unified Core Hub:** 6 shared master entities ensuring single-source-of-truth customer, employee, supplier, location, product, and account reference data.
2. **Strict Regulatory Isolation:** Native Row-Level Security (RLS) enforcement protecting Central Bank regulated VFS financial records and enforcing customer consent flags under Nigerian Data Protection laws.
3. **Separation of OLTP & OLAP:** Live transactional integrity on Supabase with zero analytical query lag due to BigQuery columnar warehouse mirroring.
4. **Actionable Executive Intelligence:** 4 targeted stakeholder scenario dashboards delivering immediate clarity to store managers, loan officers, logistics dispatchers, and group executives.

Report compiled and certified by Samuel Enyi, Lead Database Architect & SQL Engineer.